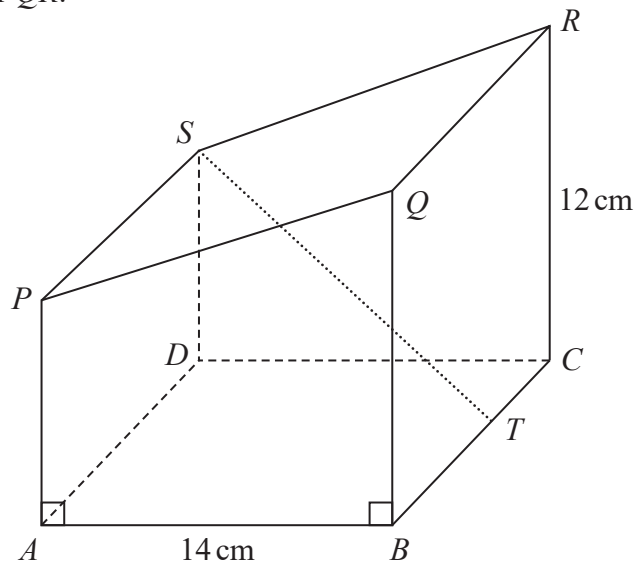


18 Here is a prism $ABCDSPQR$.



The base $ABCD$ of the prism is a square of side 14 cm
 T is the point on BC such that $BT : TC = 4 : 3$

The cross section of the prism is in the shape of a trapezium of area 147 cm^2
 $CR = 12\text{ cm}$

Find the size of the angle between the line ST and the base $ABCD$.
 Give your answer correct to 1 decimal place.